

17 A student set up an experiment to investigate diffusion.

A block of agar, $1.0\text{ cm} \times 1.0\text{ cm} \times 1.0\text{ cm}$, was stained uniformly with a water-soluble blue dye. The block of agar was put into a test-tube containing 10 cm^3 of distilled water at 20°C .

The intensity of the blue colour of the water after five minutes was measured.

Four other experiments, **A**, **B**, **C** and **D**, were then carried out using different numbers of agar blocks, different sizes of agar blocks and different temperatures. All other variables were standardised.

Which experiment would give a lighter blue colour in the water after five minutes compared to the first experiment?

	size of each agar block / cm	number of agar blocks	temperature of distilled water / $^\circ\text{C}$
A	$0.5 \times 0.5 \times 0.5$	8	30
B	$0.5 \times 0.5 \times 0.5$	2	20
C	$0.5 \times 1.0 \times 1.0$	2	30
D	$2.0 \times 1.0 \times 1.0$	1	20